

How to Drive Drivers Nuts

Effect of Auditory, Vibrotactile, and Multimodal Warnings on Perceived Urgency, Annoyance, and Acceptability

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With semiautonomous vehicles and innovative assistance systems flooding the market, the characteristics of warning signals for automotive human-machine interfaces determine their effectiveness and usage. Within the context of advanced driver assistance systems, poorly designed warning signals have the potential to startle and distract drivers. Previous studies have investigated the effectiveness of multimodal warning signals and found that such signals reduce braking times. They are also associated with high levels of perceived urgency. This study aimed to investigate the effects of auditory, vibrotactile, and multimodal warning signals on perceived urgency, annoyance, and acceptability while varying the level of participants' mental workload and the characteristics of the warning signals (interpulse interval). Results show that, compared with other signal modalities, multimodal warning signals appear to be effective in low-workload conditions but such effect disappears as the overall level of workload increases. These results are useful for the development of an adaptive warning signal system.

As semiautonomous, SAE Level 2 vehicles (1) become part of our everyday lives, the way such systems interact with the user has a significant impact on driving and, more important, safety. The user has to be kept aware of the vehicle's behavior and its surroundings at all times, especially whenever an automated-to-manual transition of control is scheduled to occur. In such instances, it is the primary task of the vehicle's human-machine interface to inform the user about the driving context and then instruct him or her about how to take back control of the vehicle in an effective and safe manner.

One of the main issues associated with semiautonomous vehicles concerns warnings and the extent to which warning signals presented by advanced driver assistance systems (ADAS) are effective in assisting drivers in regaining control of the vehicle whenever necessary. Even though it is estimated that ADAS may reduce road accidents by up to 25% (2), recent studies suggest that poorly designed warning messages may give rise to a number of unintended consequences,

including potential detrimental effects on safety. In the study by Rossi et al., for instance, authors investigated the effect of auditory warning signals on drivers' behavior and safety (3). In their study, participants drove a simulated vehicle and approached a roundabout. Whenever the recorded speed exceeded the limit, auditory signals were presented. Although such warnings were successful in reducing drivers' speed, a more fine-grained analysis conducted by Biondi et al. indicated that such an effect was instead the consequence of a startle reaction produced by the abrupt onset of the warning signal (4). The unexpected presentation of the auditory signal led drivers to suddenly release their foot from the throttle and reduce their grip on the steering wheel, two aspects with potential adverse consequences on safety. Similar results were obtained in the study by Adell et al. in which the presentation of auditory warning signals emitted by assistance systems was associated with high levels of annoyance and frustration (5).

To mitigate the unintended consequences of unimodal, auditory signals (4, 5), recent studies have investigated the effect of adaptive, multimodal warnings on aspects such as perceived urgency and annoyance. Multimodal warning signals are generated by the concurrent presentation of unimodal signals. Because of the redundancy gain effect generated by the joint processing of signals presented in different sensorial channels, multimodal signals are shown to produce faster response times than unimodal signals within highly controlled, laboratory settings (6). In the study by Biondi et al., participants drove a simulated vehicle equipped with a forward collision avoidance system; as the distance between the lead vehicle and that driven by participants fell under a safety threshold, unimodal auditory, vibrotactile, or a combination of the two signals (i.e., multimodal signals) were presented to drivers (7). In the two experiments conducted by Biondi et al., authors manipulated the level of drivers' workload by means of having them, respectively, talk on a hands-free cell phone and drive in dense traffic. Results showed that, compared with unimodal auditory and vibrotactile signals, braking times to multimodal signals were shorter with participants both talking on a cell phone and driving in high density traffic. In addition, multimodal signals produced high levels of perceived urgency, suggesting that employing such signals may be adequate in high emergency situations, rather than in most common, less urgent conditions.

In the automotive domain, adaptive human-machine interfaces are those whose way of communicating to drivers varies depending on the characteristics of the user (e.g., age, personal preferences) and the context (e.g., hazard types, weather) (8, 9). The assumption behind designing such interfaces is that, if the information provided to drivers is presented in an adaptive, tailored fashion to their

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characteristics, the level of acceptability as well as the effectiveness of such technology will be enhanced (10).

Although a number of studies have investigated the effect of multimodal warning signals in different driving and not-driving contexts (11, 12), it is not clear how multimodal warning signals affect additional aspects of subjective workload such as annoyance and acceptability. It is indeed hypothesized that high levels of annoyance associated with warning signals may reduce the level of user acceptability of the system and, as a result, its usage. In addition, this study aims to investigate how users interact with multimodal warning signals in high workload conditions. Strayer et al. (13) suggest that the level of mental workload associated with interacting with voice commands is higher than that of talking on a cell phone and similar to that of executing high cognitive demanding tasks such as the *n*-back task (14). Because responses to multimodal warning signals have been investigated in conjunction with a concurrent cell phone conversation task (7), this study aimed to investigate additional aspects of warning signals, such as acceptability, perceived urgency, and annoyance, with participants executing high cognitive demanding tasks.

EXPERIMENT 1

Experiment 1 investigated the effect of unimodal visual, unimodal auditory, unimodal vibrotactile, and multimodal signals (auditory plus vibrotactile) on subjective ratings of perceived urgency, acceptability, and frustration. In addition, the level of mental workload was manipulated with participants executing a secondary, cognitively demanding task.

Method

Participants

Eighteen students at the University of Padova (eight females) participated in this experiment. They had an average age of 23 years with a standard deviation of 3 years. They all held a valid driver license for at least 1 year. They did not report having any sight or hearing impairment.

Materials and Design

A 2×4 within-subject experimental design was considered. The factor task had two levels: single task and dual task. In the single-task conditions, participants responded to warning signals only. In the dual-task condition, in addition to responding to signals, they carried out a cognitive demanding task, the 1-back task (14, 15). The factor warning had four levels: visual, auditory, vibrotactile, and multimodal. Auditory warning signals were generated by Sound Forge 4.5 software and presented by Logitech X140 speakers. They had an amplitude of 75 dB and a frequency of 900 Hz. Vibrotactile warning signals were presented by two Lily Pad motors (20 mm in diameter), each strapped onto each participant's hand. They had an amplitude of 0.5 G. Visual warning signals were red circles with a 13-mm diameter presented over a white background on a screen located before participants. E-Prime 2.0 (Psychology Software Tools) was used to design and run the experiment.

E-Prime is a comprehensive suite of applications offering audited millisecond-timing precision, enabling researchers to develop a wide variety of paradigms that can be implemented with randomized or fixed presentation of text, pictures, and sounds (16). Eight experimental blocks were considered: four single-task and four dual-task blocks. In each block, one specific warning signal modality was presented: visual, auditory, vibrotactile, and multimodal (vibrotactile plus auditory). For each block, the same signal was presented 20 times. To avoid response anticipation, six different intertrial intervals were considered and randomly selected within each block: 600 ms, 900 ms, 1,200 ms, 1,500 ms, 1,800 ms, and 2,100 ms. To measure the level of workload associated with responding to warning signals, an augmented version of the NASA-TLX scale (17), composed of eight 21-point Likert scales, was used. Scales measured mental workload, physical workload, temporal workload, effort, performance, annoyance, perceived urgency and acceptability of the warning signals. Further information about the experimental setting, the augmented version of the NASA-TLX, and the E-Prime script may be found at <https://drive.google.com/file/d/0B5QFL8zmQG78cWdJaWExcHdPY2M/view>.

Procedure

The experiment took place in a soundproof laboratory environment and lasted for approximately 30 min. Eight experimental blocks were considered: four single-task and four dual-task blocks. The presentation of the eight blocks was counterbalanced across participants to minimize potential confounds. Within the blocks, the intertrial interval between signals was randomly selected among six possible intervals (see section on materials and design). In the single-task blocks, participants sat in front of a computer screen and were presented with the four different types of signals, one type per block. Their task was to press a button in response to the onset of the signal. In the dual-task blocks, in addition to this task, they carried out the 1-back task in which digits were presented auditorily every 2.5 s. The 1-back version of the *n*-back task (15) requires participants to listen to a series of digits presented auditorily and speak out loud the second to last digit they are presented. At the end of each 20-trial block, participants completed the augmented version of the NASA-TLX. Each block lasted approximately 1 min.

Results

A task (two levels: single task, dual task) $\times$ warnings (four levels: visual, auditory, vibrotactile, multimodal) within-subject analysis of variance (ANOVA) was considered to analyze response latency data. The analysis revealed significant main effects of task [$F(1, 17) = 19.08$, $p < .001$, partial $\eta^2 = .53$] and warnings [$F(3, 51) = 4.52$, $p < .05$, $\eta^2 = .21$]. No significant interaction was found. As expected, response times were shorter in the single-task condition compared with the dual-task condition. In the single-task condition, multimodal and vibrotactile warning signals produced faster response times compared with auditory and visual warning signals ($p < .05$). No significant difference across warning signals was found in the dual-task condition (Figure 1).

A within-subject multivariate ANOVA with tasks (two levels: single task, dual task) and warnings (four levels: visual, auditory,

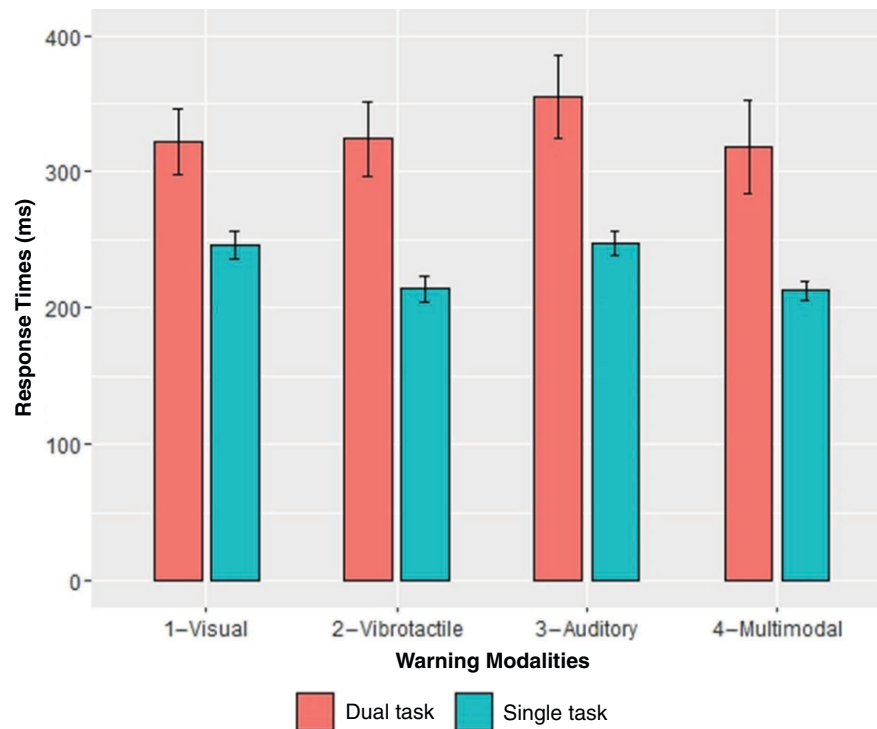


FIGURE 1 Response times to warning signals in milliseconds across single- and dual-task conditions. Error bars represent standard errors.

vibrotactile, multimodal) as the independent variable and the eight scales of the augmented NASA-TLX (mental, physical, temporal, effort, performance, annoyance, perceived urgency, acceptability) as the dependent variables was considered. Significant effects of task were found for mental workload [$F(1, 135) = 36.09, p < .05$, partial $\eta^2 = .21$], annoyance [$F(1, 135) = 23.44, p < .05$, partial $\eta^2 = .15$], and perceived urgency [$F(1, 135) = 9.43, p < .05$, partial $\eta^2 = .06$], with responding to warning signals in the dual-task condition being associated with higher ratings of mental workload ($M = 13.64$), annoyance ($M = 12.10$), and perceived urgency ($M = 12.56$) compared with those in the single-task condition (respectively, $M = 9.6, M = 8.00, M = 10.30$) ($p < .05$). Annoyance did not differ across warning signals but perceived urgency ratings for multimodal signals were significantly higher than those for other warnings [$F(3, 135) = 6.81, p < .05$, partial $\eta^2 = .13$]. Ratings for acceptability did not differ across warnings.

Overall data suggest divergent effects in the single-task versus the dual-task condition. Whereas the data collected in the single-task condition for multimodal warning signals are in agreement with the existing literature (i.e., multimodal warnings produce faster responses and higher ratings of urgency), it seems that this is not the case in the dual-task condition. The cognitive load associated with such a highly cognitive demanding task as the 1-back task zeroed the differences in response times between warning signals, making responses to multimodal warnings as fast as those to other warnings. Nonetheless, higher ratings of urgency and annoyance were associated with warnings in the dual-task condition. These represent initial evidence suggesting that the benefits associated with multimodal warnings might only hold in low workload conditions; as the work-

load increases, using multimodal warnings not only appears to fail to reduce response latencies but increases subjective ratings of annoyance and urgency. No significant difference in the ratings of acceptability was found across either task or warning conditions, suggesting that this aspect of subjective workload was not affected by these factors.

EXPERIMENT 2

Because time intervals of warning signals vary across different assistance systems (18), this experiment aimed to investigate the effect of different interpulse intervals on subjective workload, perceived urgency, and acceptability. For this reason, the only dependent variable considered in Experiment 2 is subjective workload, measured by the augmented version of the NASA-TLX. Further, given the results found for visual signals, only auditory, vibrotactile, and multimodal warnings are considered in Experiment 2.

Method

Participants

Twenty-six students at the University of Padova (16 females) participated in this experiment. They had an average age of 20 years with a standard deviation of 1.5 years. They all held a valid driver license for at least 1 year. They did not report having any sight or hearing impairment.

Materials, Design, and Procedure

A task (two levels) $\times$ warnings (three levels: auditory, vibrotactile, multimodal) $\times$ interpulse interval (IPI) (two levels: 200 ms, 600 ms) within-subject experimental design was considered. Materials used in Experiment 2 were the same as those used in Experiment 1 with the exception of IPI. With an IPI of 200 ms, one warning signal (200 ms of duration) was presented repeatedly for 10 s with an off-interval of 200 ms. With an IPI of 600 ms, the off-interval between signals was 600 ms. Twelve experimental blocks with one trial each were considered: 2 (task) $\times$ 3 (warnings) $\times$ 2 (IPI). The presentation of the 12 blocks was counterbalanced across participants to minimize potential confounds.

The procedure used in this experiment was similar to that used in Experiment 1, except that in Experiment 2 no response times were recorded. Participants listened to each signal presented repeatedly for 10 s and completed the augmented version of the NASA-TLX.

Results

Data are presented in Table 1. A within-subject multivariate ANOVA with task (two levels: single task, dual task), warnings (three levels: auditory, vibrotactile, multimodal), and IPI (two levels: 200 ms, 600 ms) as independent variables and the eight scales of the augmented NASA-TLX (mental, physical, temporal, effort, performance, annoyance, perceived urgency, acceptability) as dependent variables was considered. Significant effects of task were found for mental workload [$F(1, 300) = 305.91, p < .05$, partial $\eta^2 = .50$], annoyance [$F(1, 300) = 8.29, p < .05$, partial $\eta^2 = .03$], and acceptability [$F(1, 300) = 4.7, p < .05$, partial $\eta^2 = .01$], with ratings of mental workload and annoyance being higher in the dual-task condition compared with the single-task condition and ratings of acceptability being lower in the dual-task condition compared with the single-task condition. Significant effects of warnings were found for mental workload [$F(2, 300) =$

4.01, $p < .05$, partial $\eta^2 = .03$], annoyance [$F(2, 300) = 10.21, p < .05$, partial $\eta^2 = .06$], and perceived urgency [$F(2, 300) = 19.20, p < .05$, partial $\eta^2 = .14$], with ratings for multimodal warnings being higher than those for other warnings. Significant effects of IPI were found for mental workload [$F(1, 300) = 5.90, p < .05$, partial $\eta^2 = .01$], annoyance [$F(1, 300) = 6.5, p < .05$, partial $\eta^2 = .02$], and urgency [$F(1, 300) = 33.35, p < .05$, partial $\eta^2 = .10$], with ratings for an IPI of 200 ms being higher than those for an IPI of 600 ms. Annoyance ratings were higher in the dual-task condition compared with the single-task condition ($p < .05$), for multimodal warnings compared with other signals ($p < .05$), and with an IPI of 200 ms compared with an IPI of 600 ms ($p < .05$). The significant task times IPI interaction showed that, in the single-task condition, ratings of annoyance were higher with an IPI of 200 ms ($M = 9.41$) compared with an IPI of 600 ms ($M = 7.06$); nonetheless, no significant difference was found between the two levels of IPI in the dual-task condition (respectively, $M = 10.29$ and $M = 9.60$). Similarly, in the single-task condition, ratings of perceived urgency were higher with an IPI of 200 ms ($M = 14.29$) compared with those for an IPI of 600 ms ($M = 10.16$). However, no significant difference was found in the dual-task condition.

These results are in agreement with those of Experiment 1 so that the levels of perceived urgency and annoyance in the dual-task condition appear to be higher than those recorded in the single-task condition. Furthermore, levels of acceptability were lower in the dual-task condition, suggesting that responding to warnings in high cognitive workload conditions is less acceptable than responding to warnings in low workload conditions.

DISCUSSION OF RESULTS

The benefits of multimodal warning signals (7) might not hold in high workload conditions with drivers executing highly cognitive demanding tasks similar to that considered in Experiments 1 and 2.

TABLE 1 Mean and Standard Error of Subjective Ratings for Mental Workload, Annoyance, Urgency, and Acceptability Scales Across Warnings, Interpulse Intervals, and Task Conditions

Variable	IPI 200 ms						IPI 600 ms					
	Auditory		Vibrotactile		Multimodal		Auditory		Vibrotactile		Multimodal	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Mental												
ST	4.92	0.80	7.62	1.10	6.62	0.99	4.12	0.66	6.23	0.76	4.62	0.76
DT	13.42	0.75	14.23	0.65	14.35	0.70	13.04	0.79	13.77	0.66	13.31	0.74
Annoyance												
ST	7.00	0.89	10.65	1.03	10.58	1.16	6.08	1.09	7.96	1.05	7.15	0.74
DT	7.65	0.86	11.62	1.14	11.62	1.19	8.12	1.06	10.81	1.03	9.88	1.02
Urgency												
ST	11.77	0.90	16.27	0.67	14.85	0.69	8.62	0.88	12.62	0.96	9.27	0.82
DT	10.50	0.96	14.54	0.77	14.58	0.81	10.31	1.00	12.50	0.88	12.15	0.82
Acceptability												
ST	13.92	1.10	11.85	1.00	12.50	0.93	13.27	1.09	13.19	1.01	13.73	0.86
DT	12.73	1.04	10.92	0.98	11.15	1.12	11.31	1.14	12.12	1.00	12.50	0.9

NOTE: ST = single task; DT = dual task.

In the study by Biondi et al., participants drove a simulated vehicle and responded to the onset of multimodal warning signals presented by a forward avoidance collision system (7). When participants were talking on a cell phone, responses to multimodal signals were faster than those to vibrotactile and auditory signals. This pattern did not hold true in Experiment 1 with no difference in response times across warnings when participants executed the concurrent 1-back task, possibly because of the higher level of mental workload associated with this task compared with the cell phone conversation (13).

A relevant finding concerns the higher levels of perceived urgency and annoyance recorded in the dual-task condition compared with the single-task condition. In Experiment 1, ratings of annoyance and perceived urgency were higher in the dual-task condition. Similarly, in Experiment 2, ratings for the same scales were higher with participants executing the 1-back task and, further, no significant differences were observed between the two levels of IPI in the dual-task blocks. Instead, in the single-task condition, ratings of perceived urgency and annoyance were significantly higher with an IPI of 200 ms compared with those with an IPI of 600 ms. This effect may be explained as a consequence of the elevated level of mental stress and arousal caused by executing the 1-back task. In the study by Matthews et al., participants executed auditory and visual working memory tasks (similar to the 1-back task) (19). Results showed that the execution of working memory tasks induced high levels of participant engagement and mental distress. As mental distress is often considered to be a cause of hypervigilance (20, 21), the authors hypothesize that increased levels of mental workload generated by the 1-back task induced a higher temporary sensitivity to external factors (i.e., warning signals), so that the same signals were rated as more annoying and urgent as well as less acceptable in the dual-task condition. Support to this hypothesis is provided by the literature on the effect of mental workload on the startle response. The startle response, also referred to as startle reflex (22), is an involuntary, reactive response observed as a result of the presentation of certain types of auditory signals (e.g., high volume, unexpected signals). Studies show that as the overall level of mental workload increased, the magnitude of the startle response increased as a result, suggesting that the reaction to external signals is modulated by the amount of mental resources left available to the participant (23, 24). For an investigation of the relation between warning signals and startle response in driving, see Biondi et al. (4).

This study has one main limitation: participants were not driving. Although many similar studies conducted in a controlled laboratory setting have provided findings that were then replicated in more realistic environments (25), the fact that participants were not driving may limit the immediate applicability of these findings. Further research is therefore required to investigate the effects of high workload on drivers' responses to warning signals and, therefore, safety. The authors are interested in investigating how the high levels of frustration and low levels of acceptability associated with warnings in high workload conditions may affect additional aspects as, for example, the usage of ADAS within naturalistic driving contexts. Future studies will also look at developing different types of adaptive warnings whose characteristics (e.g., modality, frequency or location within the cockpit) will be tailored to the demographics of the user and different types of hazards.

The experiments presented here question the effectiveness of multimodal warning signals in high workload situation. In particular, in addition to the absence of any benefits in response times of multimodal warning signals in the dual-task condition, such signals increased the

ratings of annoyance and perceived urgency. These findings, if replicated in more naturalistic conditions, represent further evidence for the need of adaptive warning signals systems (9) that factor the driver state and demographics into the development of effective human-machine interfaces.

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